RAVI PATEL

Software & Game Developer

Unity • Unreal Engine 5 • Three.js

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LOCATION
Anand, Gujarat, India

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PORTFOLIO
raviif.github.io

SKILLS

LANGUAGES
C# | C++ | Python | JavaScript | TypeScript | HLSL/GLSL

ENGINES & 3D
Unity | Unreal Engine 5 | Three.js | WebGL | Shaders | Blueprints | Procedural Meshes | Physics

XR
HoloLens 2 | MRTK | OpenXR | Meta Quest | XREAL | CT/MRI Viz

COMPUTER VISION & AI
MediaPipe | OpenCV | YOLO | ONNX Runtime

WEB & KIOSK
Electron | Node.js | WebSockets | REST APIs | IndexedDB | Linux Kiosk

SECURITY
E2E Encryption | X25519 | HKDF-SHA256 | AES-256-GCM

TOOLS
Git/GitHub | Visual Studio | Rider | Linux | Windows

EDUCATION

B.E. in Computer Engineering
Ipcowala Institute of Engineering & Technology, Anand, Gujarat
Graduated 2023 | CGPA 8.30 / 10

CERTIFICATIONS

Emerging Technologies: AI, IoT & ERP
Code Unnati (SAP CSR) | Edunet Foundation, 2022–23

Learn the ABCs of ABAP Programming on SAP BTP
SAP

Claude 101
Claude Code 101
Claude Platform 101
Anthropic

PROFILE

Software and Game Developer with 3+ years building real-time 3D, XR, and computer-vision applications in Unity, Unreal Engine 5, and Three.js. Solo-built **SmartScopy**, an XR platform that streams live endoscopic surgery to XREAL glasses and HoloLens 2. Built the encryption, kiosk, and speech-therapy layers of **NoPainHeals**, a commercial therapy platform, plus **MimadimPlus**, an automated CAD-to-3D architectural engine in Unreal Engine 5.

PROFESSIONAL EXPERIENCE

Software & Game Developer | Invisible Fiction, Anand, Gujarat — May 2023 – Present

SmartScopy — Live Operating Theatre XR Platform

Unity, C#, XREAL, HoloLens 2, MRTK/OpenXR

- Independently designed and built an XR platform that streams live endoscopic surgery camera feeds in real time to XREAL glasses and HoloLens 2, so the surgical team can follow the procedure directly in the headset.
- Developed the low-latency live video pipeline in Unity (C#) for real-time display of operating theatre camera feeds.
- Integrated interactive 3D anatomical models reconstructed from CT, MRI, and X-ray scans, viewable alongside the live surgical feed.

NoPainHeals — Multi-Modal Therapy Platform

Electron, Three.js, WebGL, WebSockets, Linux Kiosk

- Core contributor to a clinical therapy platform (Three.js, with AI-assisted development) spanning motor therapy, body-tracking games, floor-projection rehabilitation, balance therapy, and an AI speech-therapy station. **Owned** the encryption pipeline, kiosk deployment, and speech-therapy station.
- Delivered the floor-projection rehabilitation suite with **12+ interactive games** that improve mobility and cognitive reaction.
- Deployed and secured the commercial kiosk app across Linux and Windows: system lockdown, hardware identification, auto-launch controls.
- Designed an end-to-end encrypted pipeline for confidential patient telemetry, plus an offline-first queue with automatic retry that prevents data loss during clinic network outages.
- Integrated real-time WebSocket communication for cloud updates, live remote control, and device pairing.
- Refactored a large legacy WebGL codebase into a modular 3D architecture; integrated AI speech recognition for automated articulation scoring and real-time audio feedback for pediatric patients.

MimadimPlus — Automated 3D Architectural Visualization

Unreal Engine 5, C++, Python

- Built an automated 2D-to-3D engine that converts raw CAD/DXF blueprints into realistic, real-time 3D building environments, eliminating manual modeling effort.
- Developed procedural algorithms for walls, floors, roofs, doors, and windows with accurate lighting, materials, and textures, plus dynamic flooring for multi-room layouts.
- Created spatial-transformation and room-detection logic for blueprint alignment, layout extraction, and fixture placement.
- Built a companion Python desktop app to parse CAD files and stream blueprint data into Unreal; supported 2D layout editing and Meta Quest VR walkthroughs.

Camera-Based Motion Tracking & Edge AI Systems

Python, OpenCV, MediaPipe, YOLO, ONNX, Unity

- Deployed custom motion and object-tracking models on live camera input, optimized for real-time performance on standard hardware.
- Designed a hybrid computer-vision pipeline for body pose and gesture recognition without specialized sensors.
- Built controller-free games (Fruit Samurai, Alien Runner, Enter If You Dare, Football Cup) that turn full-body movement into gameplay.

Computer Engineering Intern | Invisible Fiction — Jan 2023 – May 2023

- Developed an Iris Recognition and Distance Detection prototype in Python for user-to-screen distance tracking and biometric identification.

ADDITIONAL PROJECTS

- **3D Bone Visualizer** *(Unity, HoloLens 2, MRTK, OpenXR)* — mixed-reality app where surgeons inspect, rotate, slice, and manipulate 3D bone models from CT/MRI/X-ray data using hand-tracked gestures.
- **MimadimHomeStudio** *(UE5, C++, Blueprints, Data Assets)* — real-time interior customization of furniture, materials, colors, and lighting, with dynamic furniture swapping that preserves position and rotation.
- **VR Boat Simulation** *(Unity, C#, VR)* — maritime training sim helping pilots move from small vessels to cargo ships; built water effects and boat physics.
- **VR Zombie Game & Clean the City** *(Unity, C#)* — a first-person VR shooter, and a mobile game where players control a black hole to collect debris.
- **Creative Spaces Website** *(Freelance; HTML5, CSS3, JavaScript, AI-assisted)* — responsive static website built for an interior design client.